

Solving $Ax = b$: Row Reduced Form R

Now the client wants a specific target. Four short steps on one simplified matrix $[R \mid d]$ tell you whether it can be hit, give one portfolio that hits it, and then every portfolio that does.

PART 1 Index

One question: **which portfolios hit a client's target?** Four short steps, all done on one simplified matrix $[R \mid d]$.

1. The question, and the plan

The four-fund menu, a target, and the four steps.

2. Step 1: from $[A \mid b]$ to $[R \mid d]$

Lesson 7's row operations, also applied to the target column.

3. Step 2: what does the zero row tell us?

" $0 = 0$ " is fine; " $0 = -2$ " means no portfolio can hit the target.

4. Step 3: one answer

Free amounts = 0; the pivot amounts are the last column.

5. Step 4: every answer

Pick any free amounts and R gives the rest: one answer + any do-nothing trade. With a target tester.

6. The same four steps on smaller menus

Three 2×2 s (one, none, infinitely many) and two 3×3 s.

7. How many answers? The rank tells you

No zero rows $\Rightarrow$ always reachable. No free columns $\Rightarrow$ never more than one.

METHODS IN THIS LESSON

[Augmented matrix \$\[A \mid b\]\$](#)
[Reduce to \$\[R \mid d\]\$](#)
[Zero-row test](#)
[Particular solution](#)
[Complete solution \$x_p + x_n\$](#)
[Rank cases](#)

PART 2 Prerequisites

Click any link to review it, then come back.

- From U to R, pivot and free variables and special solutions (Lesson 7).
- $Ax = b$ is solvable exactly when b is in $C(A)$ (Lesson 6).

- Elimination with the right side (Lesson 2).

PART 3 Explanation

THE WHOLE LESSON IN ONE LINE

Simplify $[A \mid b]$ to $[R \mid d]$. A zero row with a non-zero end means **no answer**. Otherwise set the free amounts to 0 and read **one answer** off d . Add any do-nothing trade to get **every answer**.

1. The question, and the plan

Lesson 7 asked which trades change **nothing** ($Ax = 0$). Now a client wants something specific: a target b . Same menu as Lesson 7:

THE MENU AND THE TARGET

Funds: Nifty $(1, 12, 1)$, Nifty ETF $(1, 12, 1)$, bonds $(1, 7, 0)$, balanced $(1, 9.5, 0.5)$. Amounts $x = (x_1, x_2, x_3, x_4)$ in ₹ lakh.

The client wants budget ₹10 lakh, return ₹105 thousand, exposure 7: $b = (10, 105, 7)$.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \\ 7 \end{bmatrix}$$

Three things can happen: the target is **impossible**, there is **exactly one** portfolio, or there are **infinitely many**. The plan finds out which, and gives every portfolio:

Step	What you do	What it tells you
1	Simplify $[A \mid b]$ to $[R \mid d]$	the simplest form of the rules and the target
2	Look at the zero rows	is the target possible?
3	Set free amounts to 0, read d	one portfolio that works
4	Add the do-nothing trades	every portfolio that works

2. Step 1: from $[A \mid b]$ to $[R \mid d]$

METHOD · THE AUGMENTED MATRIX $[A \mid B]$

Write the target as an extra column on the right. Every row operation is applied to the whole row, including that last entry. Each row still means one rule: "these amounts give this number".

$$[A \mid b] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 10 \\ 12 & 12 & 7 & 9.5 & 105 \\ 1 & 1 & 0 & 0.5 & 7 \end{array} \right]$$

Now do **exactly the five row operations of Lesson 7**. The first four columns change exactly as they did there; the only new work is the last column.

1 ROW 2 $- 12 \times$ ROW 1 $(0, 0, -5, -2.5 \mid 105 - 12(10)) = (0, 0, -5, -2.5 \mid -15)$.
MEANING "Return beyond 12%" must be -15 : the client wants ₹15 thousand less than an all-Nifty ₹10 lakh would earn.

2 ROW 3 $-$ ROW 1 $(0, 0, -1, -0.5 \mid 7 - 10) = (0, 0, -1, -0.5 \mid -3)$.

3 ROW 3 $- 0.2 \times$ ROW 2 $(0, 0, 0, 0 \mid -3 - 0.2(-15)) = (0, 0, 0, 0 \mid -3 + 3) = (0, 0, 0, 0 \mid 0)$.

4 ROW 2 $\div (-5)$ $(0, 0, 1, 0.5 \mid -15 / -5) = (0, 0, 1, 0.5 \mid 3)$.

5 ROW 1 $-$ ROW 2 $(1, 1, 0, 0.5 \mid 10 - 3) = (1, 1, 0, 0.5 \mid 7)$.

METHOD · REDUCE $[A \mid B]$ TO $[R \mid D]$

Clear below the pivots ($A \rightarrow U$), then make each pivot 1 and clear above it ($U \rightarrow R$), all on the full rows. The left part becomes R (the same R as for $Ax = 0$); the last column becomes d .

$$[R | d] = \left[\begin{array}{cccc|c} \boxed{1} & 1 & 0 & 0.5 & 7 \\ 0 & 0 & \boxed{1} & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + 0.5x_4 = 7 \\ \text{row 2: } x_3 + 0.5x_4 = 3 \\ \text{row 3: } 0 = 0 \end{array}$$

Everything we need is in this one matrix.

IN ONE LINE

Same row operations as Lesson 7, carried through the target column too.

3. Step 2: what does the zero row tell us?

We reached:

$$[R | d] = \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0.5 & 7 \\ 0 & 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Look only at the last row:

$$0 \quad 0 \quad 0 \quad 0 \quad | \quad 0$$

It says $0x_1 + 0x_2 + 0x_3 + 0x_4 = 0$. That means:

$$\boxed{0 = 0}$$

That's perfectly fine. It doesn't give us any new restriction.

But imagine instead we got:

$$0 \quad 0 \quad 0 \quad 0 \quad | \quad -2$$

That would mean:

$$\boxed{0 = -2}$$

That's impossible. No matter what values we choose for x_1, x_2, x_3, x_4 , the left side will always be zero. Therefore:

No portfolio can hit the target.

This really happens. Ask for exposure 5 instead of 7, target $(10, 105, 5)$, and the same row operations end the last row with $(5 - 10) - 0.2(105 - 120) = -5 + 3 = -2$. The last row is

exactly $0 \ 0 \ 0 \ 0 \mid -2$.

RULE · THE ZERO-ROW TEST

Every all-zero row of R must end in 0. If one ends in a non-zero number, the target is impossible. (If R has no zero rows at all, every target passes.)

So remember just this:

$$0 = 0 \Rightarrow \text{target is possible}$$

$$0 = \text{non-zero number} \Rightarrow \text{target is impossible}$$

For our target $(10, 105, 7)$ we have $0 = 0$, so the target is possible.

▼ For the curious: why exposure 5 is impossible, and the test for any target

The zero row came from the exposure rule. For these four funds, exposure is already decided once budget and return are fixed: ₹10 lakh earning ₹105 thousand forces exposure 7. Asking for 5 is asking for two different things at once.

With letters $b = (B, R, E)$, the last row ends in $\frac{1}{5}(7B + 5E - R)$. So a target is possible exactly when $R = 7B + 5E$, the plane $C(A)$ from Lesson 5.

4. Step 3: one answer

The target passed. Now look at the two rows that are left:

$$\begin{aligned}x_1 + x_2 + 0.5x_4 &= 7 \\x_3 + 0.5x_4 &= 3\end{aligned}$$

No row decides the ETF (x_2) or balanced (x_4): they are the **free** amounts (Lesson 7). The simplest choice is to put zero in them:

$$x_2 = 0, \quad x_4 = 0$$

Then the rows give the other two amounts straight away:

$$x_1 = 7 \quad x_3 = 3$$

So one portfolio is:

$$x_p = (7, 0, 3, 0)$$

₹7 lakh Nifty, ₹3 lakh bonds

Check in A : budget $7 + 0 + 3 + 0 = 10$ ✓, return $84 + 0 + 21 + 0 = 105$ ✓, exposure $7 + 0 + 0 + 0 = 7$ ✓.

METHOD · PARTICULAR SOLUTION x_p

Put 0 in every free amount. Each pivot amount is then simply the last number d of its row.

So remember just this:

$$\text{free amounts} = 0 \Rightarrow \text{pivot amounts} = d$$

5. Step 4: every answer

Zero was only the simplest choice. The free amounts can be anything. Try ETF $x_2 = 3$ and balanced $x_4 = 2$:

$$\begin{aligned} x_3 + 0.5(2) &= 3 &\Rightarrow &\boxed{x_3 = 2} \\ x_1 + 3 + 0.5(2) &= 7 &\Rightarrow &\boxed{x_1 = 3} \end{aligned}$$

So another portfolio is:

$$(3, 3, 2, 2)$$

₹3L Nifty, ₹3L ETF, ₹2L bonds, ₹2L balanced

Check in A : budget $3 + 3 + 2 + 2 = 10$ ✓, return $36 + 36 + 14 + 19 = 105$ ✓, exposure $3 + 3 + 0 + 1 = 7$ ✓.

Every choice of free amounts gives another portfolio. With ETF = c and balanced = d , the rows give Nifty = $7 - c - 0.5d$ and bonds = $3 - 0.5d$. Split that up:

$$\begin{bmatrix} 7 - c - 0.5d \\ c \\ 3 - 0.5d \\ d \end{bmatrix} = \underbrace{\begin{bmatrix} 7 \\ 0 \\ 3 \\ 0 \end{bmatrix}}_{\text{one answer (Step 3)}} + c \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\text{do-nothing trades (Lesson 7)}} + d \underbrace{\begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix}}_{\text{do-nothing trades (Lesson 7)}}$$

RULE · EVERY ANSWER = ONE ANSWER + A DO-NOTHING TRADE

Adding a trade that changes nothing can't spoil a portfolio that hits the target: $A(x_p + x_n) = b + 0 = b$. And there's nothing else: if x is any answer, then $x - x_p$ changes nothing, so it's a do-nothing trade.

In money terms: once one portfolio hits the target, you can swap Nifty for the ETF, or build balanced from Nifty and bonds, as much as you like. The target stays hit.

So remember just this:

every answer = one answer + any do-nothing trade

Try it: any target, any free amounts

Budget B Return R Exposure E

ETF c Balanced d

Zero-row test: $7B + 5E - R = 7(10) + 5(7) - 105 = 0$.

$[R \mid d]$ has last column $d = (7, 3, 0)$. **One answer** (free amounts = 0): $x_p = (7, 0, 3, 0)$.

Every answer (ETF = c , balanced = d): $x = x_p + 2s_1 + 2s_2 = (4, 2, 2, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(4) + 1(2) + 1(2) + 1(2) = 10$ ✓

Return: $12(4) + 12(2) + 7(2) + 9.5(2) = 105$ ✓

Exposure: $1(4) + 1(2) + 0(2) + 0.5(2) = 7$ ✓

Hits the target. Change c and d : it still does.

6. The same four steps on smaller menus

Same steps, smaller matrices. These show all three outcomes.

2 × 2: Nifty and bonds, target (10, 105): exactly one answer

1 STEP 1

$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 7 & 105 \end{array} \right]$. Row 2 $- 12 \times$ row 1: $(0, -5 \mid 105 - 120 = -15)$. Row 2 $\div (-5)$: $(0, 1 \mid 3)$. Row 1 $-$ row 2: $(1, 0 \mid 7)$.

STEP 2

No zero rows: possible.

STEPS 3–4

No free columns, so nothing to choose: $x = (7, 3)$ is the **only** answer.

2 × 2: Nifty and its twin, target (10, 120): infinitely many

1 STEP 1

$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 12 & 120 \end{array} \right]$. Row 2 $- 12 \times$ row 1: $(0, 0 \mid 120 - 120 = 0)$.

STEP 2

Zero row ends in 0: possible.

STEP 3

ETF free = 0: Nifty = 10, so $x_p = (10, 0)$.

STEP 4

$x = (10, 0) + c(-1, 1)$: any split of ₹10 lakh between the twins.

2 × 2: Nifty and its twin, target (10, 105): no answer

1 STEP 1

Row 2 $- 12 \times$ row 1: $(0, 0 \mid 105 - 120 = -15)$.

STEP 2

The zero row says $0 = -15$: **impossible**. Two copies of a 12% fund can't earn 10.5%.

3 × 3: Nifty, bonds, midcap, target (10, 111, 7.4): exactly one answer

1 CLEAR BELOW

Row 2 $- 12 \times$ row 1: $(0, -5, 3 \mid 111 - 120 = -9)$. Row 3 $-$ row 1: $(0, -1, 0.2 \mid 7.4 - 10 = -2.6)$. Row 3 $- 0.2 \times$ row 2: $(0, 0, -0.4 \mid -2.6 + 1.8 = -0.8)$.

CLEAR ABOVE

Row 3 $\div (-0.4)$: $(0, 0, 1 \mid 2)$. Row 2 $- 3 \times$ row 3: $(0, -5, 0 \mid -9 - 6 = -15)$, then $\div (-5)$: $(0, 1, 0 \mid 3)$. Row 1 $-$ row 2 $-$ row 3: $(1, 0, 0 \mid 10 - 3 - 2 = 5)$.

READ

$R = I$: no zero rows, no free columns. The only answer is $(5, 3, 2)$. Check: $5 + 3 + 2 = 10 \checkmark$, $60 + 21 + 30 = 111 \checkmark$, $5 + 0 + 2.4 = 7.4 \checkmark$.

3 × 3: Nifty, bonds, balanced, target (10, 105, 7): infinitely many

1 STEP 1

As in topic 2 without the ETF column: $[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$.

STEP 2

Zero row ends in 0: possible.

STEPS 3–4

Balanced free = 0: $x_p = (7, 3, 0)$. Every answer: $(7, 3, 0) + t(-0.5, -0.5, 1)$, as in Lesson 6.

▼ **Check yourself (MIT lecture):** solve $\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} x = (1, 5, 6)$.

Step 1: row 2 $- 2 \times$ row 1 $= (0, 0, 2, 4 \mid 3)$; row 3 $- 3 \times$ row 1 $= (0, 0, 2, 4 \mid 3)$; row 3 $-$ row 2 $= (0, 0, 0, 0 \mid 0)$; row 2 $\div 2 = (0, 0, 1, 2 \mid \frac{3}{2})$; row 1 $- 2 \times$ row 2 $= (1, 2, 0, -2 \mid -2)$. Step 2: zero row ends in 0 ✓. Step 3: $x_2 = x_4 = 0$ gives $x_p = (-2, 0, \frac{3}{2}, 0)$. Step 4: add $c_1(-2, 1, 0, 0) + c_2(2, 0, -2, 1)$.

▼ **Check yourself (MIT recitation):** when can $x - 2y - 2z = b_1, 2x - 5y - 4z = b_2, 4x - 9y - 8z = b_3$ be solved?

Row 2 $- 2 \times$ row 1: $(0, -1, 0 \mid b_2 - 2b_1)$. Row 3 $- 4 \times$ row 1: $(0, -1, 0 \mid b_3 - 4b_1)$. Row 3 $-$ row 2: $(0, 0, 0 \mid b_3 - b_2 - 2b_1)$. Zero-row test: solvable exactly when $b_3 - b_2 - 2b_1 = 0$. Finishing: $y = 2b_1 - b_2, x = 5b_1 - 2b_2 + 2z$, so every answer is $(5b_1 - 2b_2, 2b_1 - b_2, 0) + c(2, 0, 1)$. For $b = (1, 1, 3)$: $3 - 1 - 2 = 0$ ✓ and $x_p = (3, 1, 0)$.

IN ONE LINE

Bad zero row $\Rightarrow$ none. Otherwise: free column $\Rightarrow$ infinitely many; no free column $\Rightarrow$ exactly one.

7. How many answers? The rank tells you

Topic 6 showed the outcome depends on two things in R : zero rows and free columns. Both are counted by the rank r (the number of pivots), for a menu with m rules and n funds:

- Zero rows $= m - r$. If $r = m$ there are none, so **every target is possible**.
- Free columns $= n - r$. If $r = n$ there are none, so there is **never more than one answer**.

RULE · THE FOUR RANK CASES

So the rank tells you what can happen before you even look at the target.

Rank	Zero rows?	Free columns?	Answers	Menu
$r = m = n$	no	no	exactly 1, for every target	Nifty, bonds, midcap
$r = n < m$	yes	no	0 or 1	Nifty, bonds (3 rules)
$r = m < n$	no	yes	infinitely many, for every target	Nifty, bonds, midcap, balanced
$r < m,$ $r < n$	yes	yes	0 or infinitely many	Nifty, ETF, bonds, balanced

An example of the third row. Nifty, bonds, midcap, balanced has rank 3 = 3 rules: no zero rows. So even $(10, 105, 5)$, impossible for our main menu, is possible here, for example $(-1, 6, 5, 0)$ (short ₹1 lakh Nifty, ₹6 lakh bonds, ₹5 lakh midcap). Check: $-1 + 6 + 5 = 10 \checkmark$, $-12 + 42 + 75 = 105 \checkmark$, $-1 + 0 + 6 = 5 \checkmark$.

READING IT AS A TRADER

$r = m$: your products can hit every target you measure. $r = n$: no product is redundant, so every answer is unique. Real desks usually have more products than risks, and use the spare freedom to pick the cheapest or most liquid route.

IN ONE LINE

$r = m \Rightarrow$ always possible; $r = n \Rightarrow$ at most one answer; both $\Rightarrow$ exactly one.